

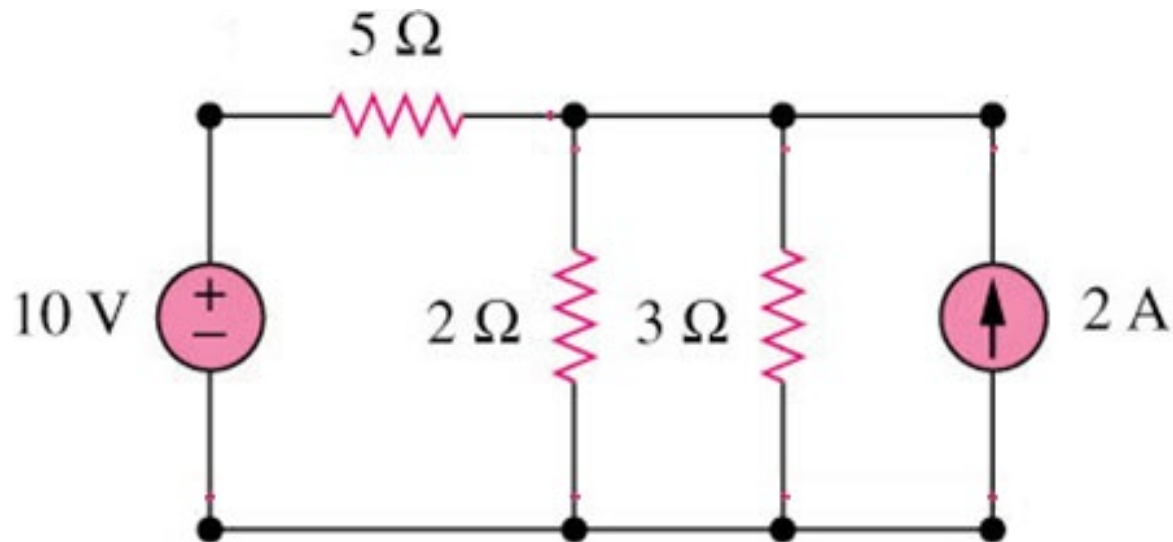
Electrical Science

ENGR 2303

Dr. Mohamed Bingabr
School of Engineering
University of Central Oklahoma

Ohm's and Kirchhoff's Laws

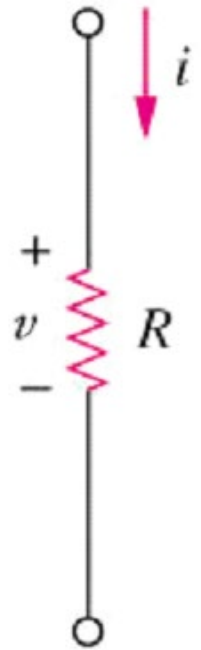
- Ohm's Law.
- Nodes, Branches, and Loops.
- Kirchhoff's Laws.



Circuit Theory Video 2

Ohm's Law

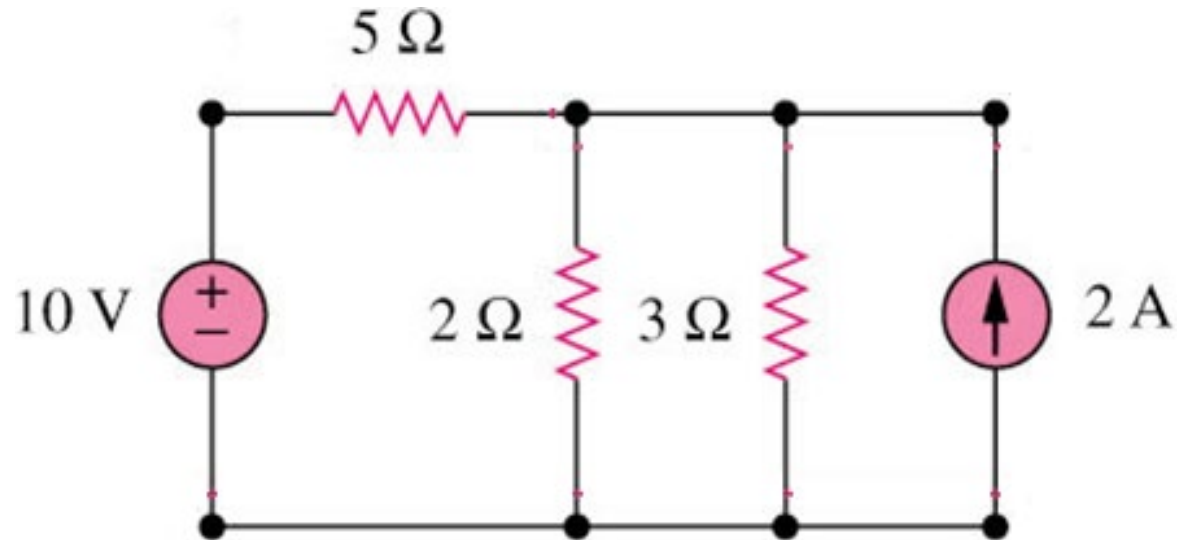
- Ohm's law $v = iR$
- Short Circuit $R = 0$
- Open Circuit $R = \infty$
- Conductance $G = 1/R$
- Power $P = iv = i^2R = v^2/R$



Nodes, Branches and Loops

- A **branch** represents a single element such as a voltage source or a resistor.
- A **node** is the point of connection between two or more branches.
- A **loop** is any closed path in a circuit.

$$b = l + n - 1$$



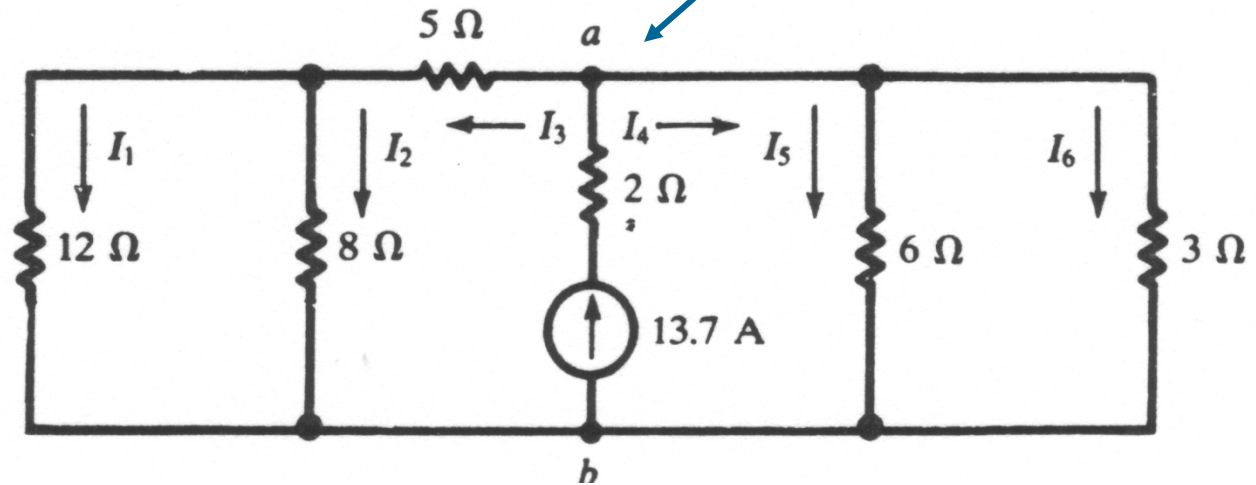
Nodes, Branches and Loops

Example

How many branches, nodes and loops are there?

$$b = l + n - 1$$

Should we consider it as one branch or two branches?

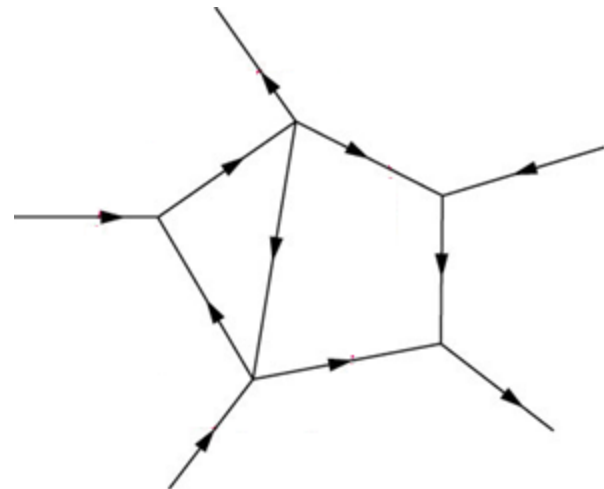
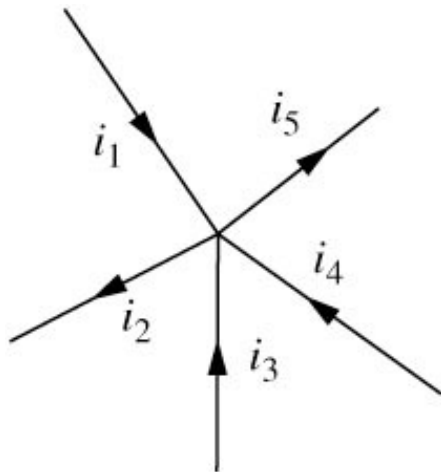


Kirchhoff's Current Law

- Kirchhoff's current law (KCL) states that the algebraic sum of currents entering and leaving a node (or a closed boundary) is zero.

Mathematically,
$$\sum_{n=1}^N i_n = 0$$

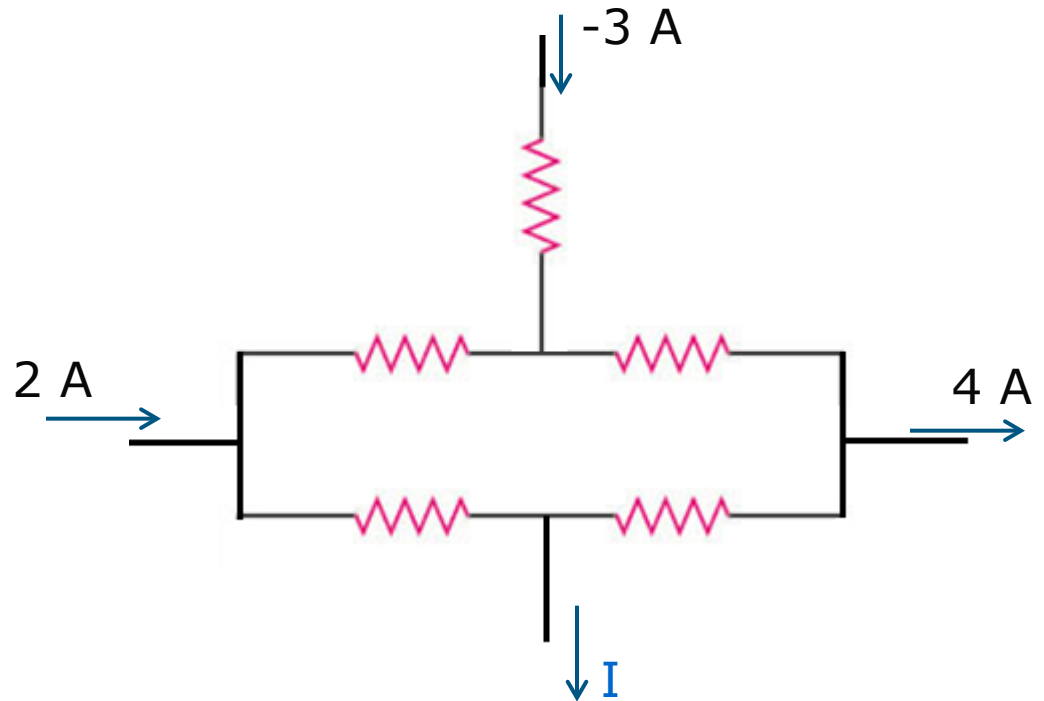
- Current entering a node is positive and leaving a node is negative.



Kirchhoff's Current Law

Example

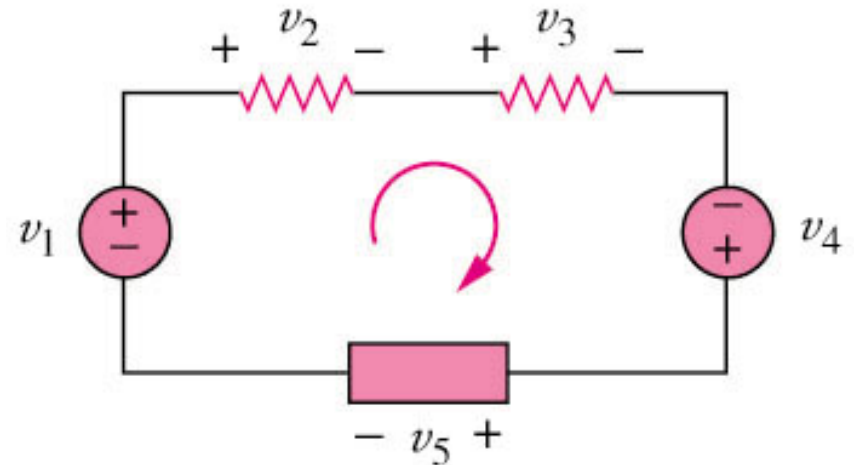
Determine the current I for the circuit shown in the figure below.



Kirchhoff's Voltage Law

- Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

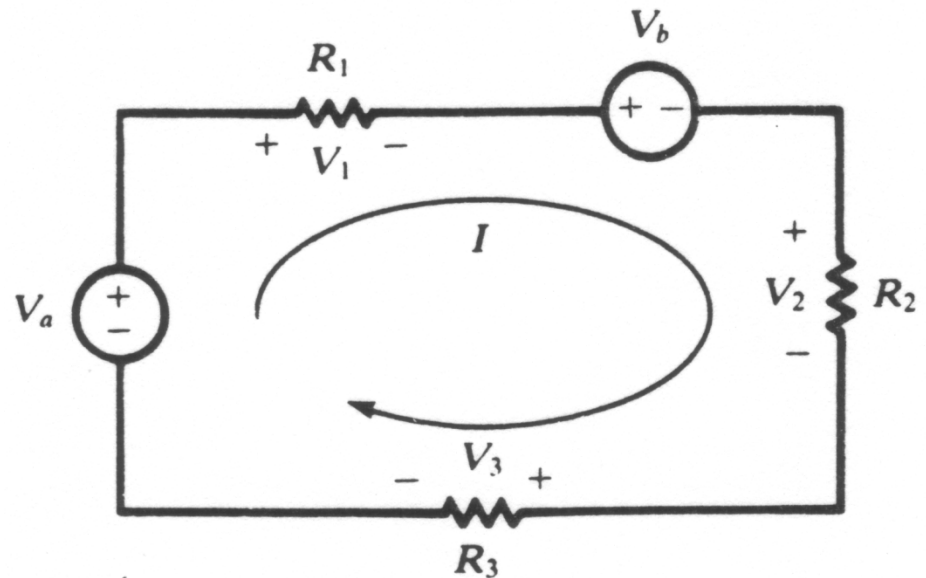
Mathematically, $\sum_{n=1}^N v_n = 0$



Kirchhoff's Voltage Law

Example

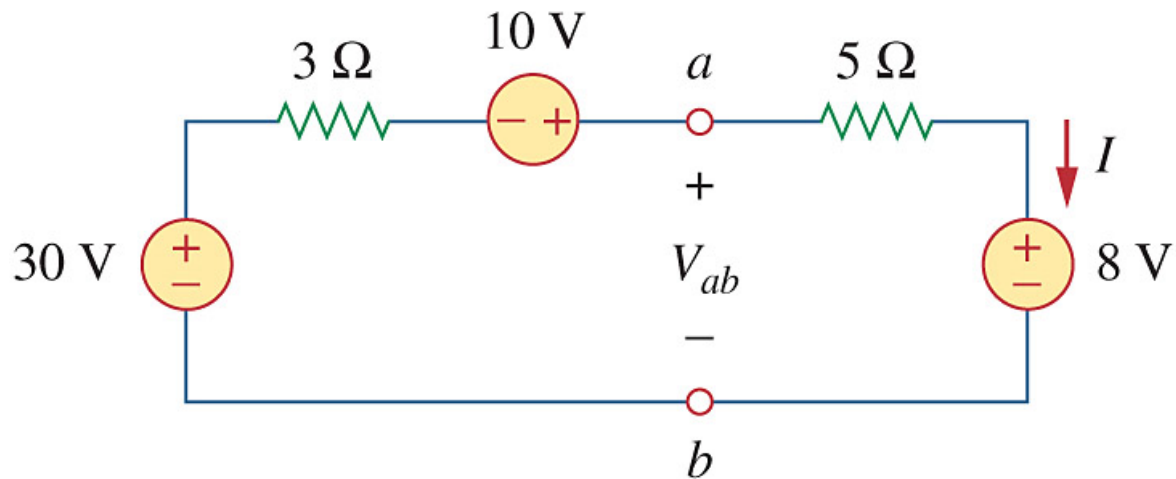
Applying the KVL equation for the circuit of the figure below. Find I .



Kirchhoff's Voltage Law

Example

Find I and V_{ab} in the circuit.



Example: Kirchhoff's Voltage Law

Example

Find V_x in the circuit.

